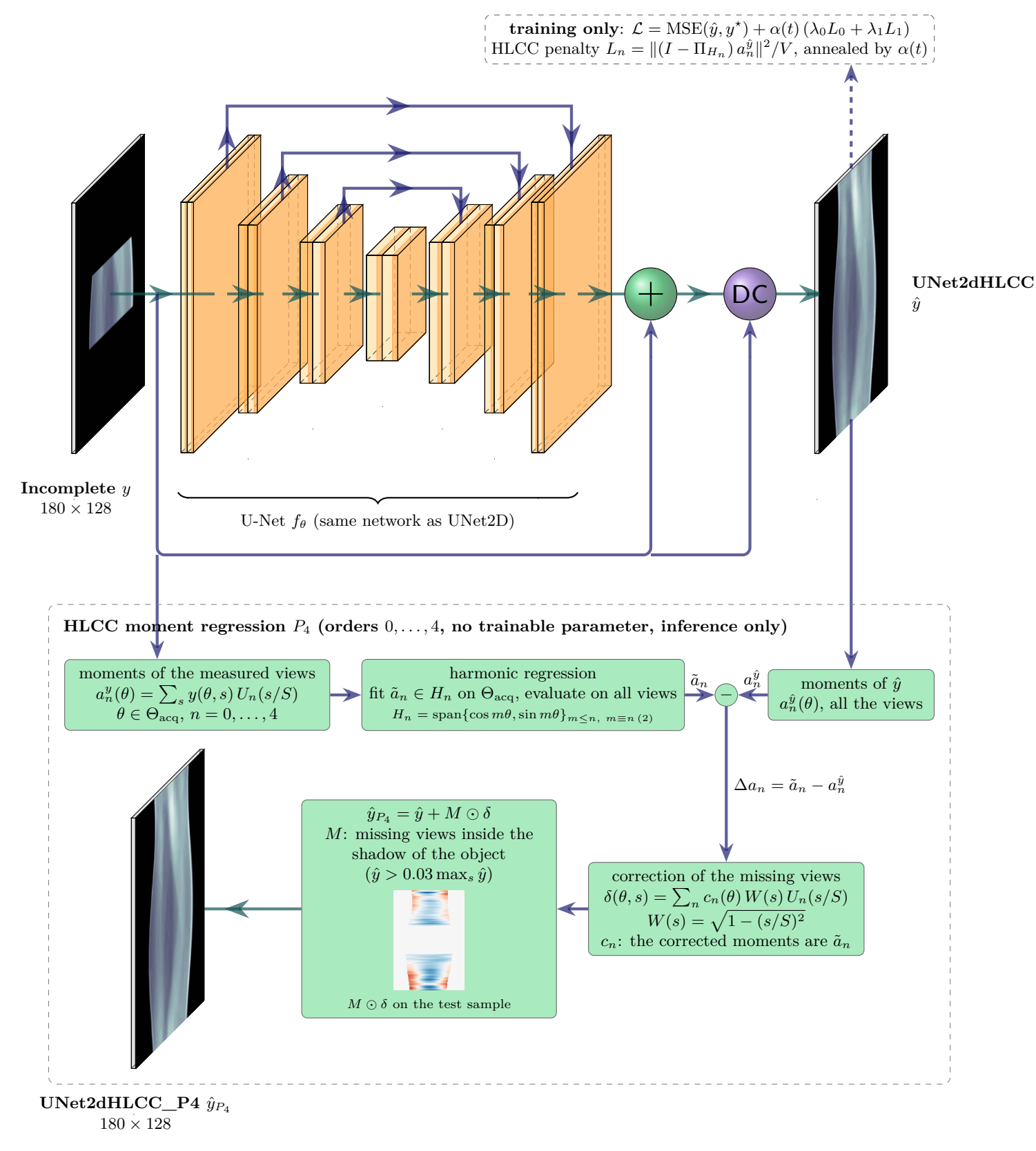

training only: $\mathcal{L} = \mathrm{MSE}(\hat{y}, y^\star) + \alpha(t)\,(\lambda_0 L_0 + \lambda_1 L_1)$
HLCC penalty $L_n = \|(I - \Pi_{H_n})\, a_n^{\hat{y}}\|^2 / V$, annealed by $\alpha(t)$
DC
UNet2dHLCC
$\hat{y}$
Incomplete y
180×128
U-Net f_θ (same network as UNet2D)
HLCC moment regression P_4 **(orders** $0, \ldots, 4$**, no trainable parameter, inference only)**
moments of the measured views
$a_n^y(\theta) = \sum_s y(\theta, s)\, U_n(s/S)$
$\theta \in \Theta_{\mathrm{acq}}$, $n = 0, \ldots, 4$
harmonic regression
fit $\tilde{a}_n \in H_n$ on Θ_{acq}, evaluate on all views
$H_n = \mathrm{span}\{\cos m\theta, \sin m\theta\}_{m \le n,\ m \equiv n\ (2)}$
$\tilde{a}_n$
$a_n^{\hat{y}}$
moments of $\hat{y}$
$a_n^{\hat{y}}(\theta)$, all the views
$\Delta a_n = \tilde{a}_n - a_n^{\hat{y}}$
correction of the missing views
$\delta(\theta, s) = \sum_n c_n(\theta)\, W(s)\, U_n(s/S)$
$W(s) = \sqrt{1 - (s/S)^2}$
c_n: the corrected moments are $\tilde{a}_n$
$\hat{y}_{P_4} = \hat{y} + M \odot \delta$
M: missing views inside the shadow of the object
$(\hat{y} > 0.03 \max_s \hat{y})$
$M \odot \delta$ on the test sample
UNet2dHLCC_P4 $\hat{y}_{P_4}$
180×128